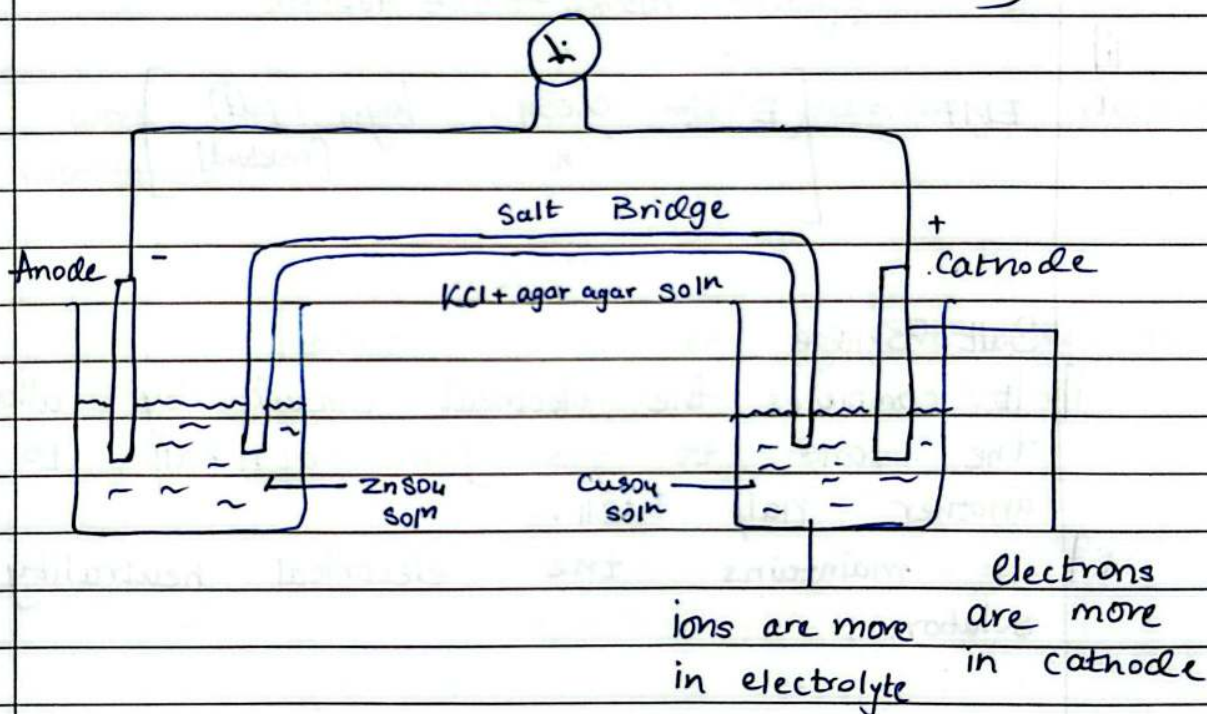


Chapter ~ 2

Electrochemistry



Standard Electrode Potential (E°)

$$EMF = E_c - E_A \quad \text{or} \quad E^\circ = E_c - E_A$$

Daniel Galvanic Cell $\Rightarrow$ Zn & Cu

$$SRP = -SOP$$

Cell representation : $Zn/Zn^{+2} // Cu^{+2}/Cu$
 $\uparrow$ Phase Boundary

All Electrode Potential values are Standard
 Standard form $\Rightarrow$ 1 molal — conc.
 1 atm — pressure
 298 K — temp.

Formulae for EMF. $\xrightarrow{8.314 \rightarrow 298 \text{ K}}$

$$i) \quad EMF = \left[E^\circ - \frac{2.303RT}{nF} \log_{10} \frac{[pdt]}{[reactant]} \right]$$

moles — 96500

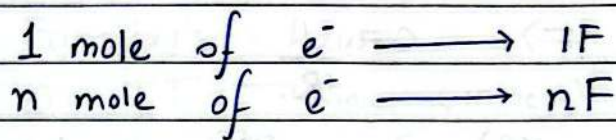
$$ii) \quad EMF = \left[E^\circ - \frac{0.059}{n} \log_{10} \frac{[pdt]}{[reactant]} \right]$$

Salt Bridge

- i) It completes the electrical circuit by allowing the ions to flow from one half to another half cell.
- ii) It maintains the electrical neutrality of solution.

$$W = q \times V \text{ — potential}$$

| Charge



$$\therefore \text{ work done } \Rightarrow W = nFE^\circ$$

(Spontaneous Reaction) — by the system.
uses Gibbs Free energy.

work increases $\Rightarrow$ Gibbs free energy decreases
(spontaneous)

$$\begin{aligned} W &= -\Delta G \\ \Delta G &= -nFE^\circ_{\text{cell}} \\ \text{At Equilibrium} \\ \Delta G &= -2.303 RT \log K_c \end{aligned} \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{-ve value in} \\ \text{both} \end{array}$$

$\Rightarrow$ Electrolytic Conductance

1) Resistance (R) $\Rightarrow V = IR.$

2) Conductance (C) $\Rightarrow C = \frac{1}{R}.$

3) Specific Resistance (ρ) $\Rightarrow \frac{RA}{l} \Rightarrow$ distance b/w electrodes

$\frac{l}{A} = \text{Cell Constant}$ or $\boxed{\frac{l}{A} = K \times R}$

4) Conductivity (K) / Specific Conductance
 $\kappa \downarrow$

i) $\Rightarrow \frac{1}{\rho} = \frac{1}{\frac{RA}{l}} = \frac{l}{RA}$

ii) $\Rightarrow$ conductance $\times$ Cell constant
 $\Rightarrow \frac{1}{R} \times \frac{l}{A}.$

5) Molar Conductivity (Δm)

i) $\Rightarrow \frac{K \times 1000}{\text{molarity}}$ or

ii) $\Rightarrow K \times \text{Volume}$

6) Equivalent Conductivity (Δeq)

$\Rightarrow \frac{K \times 1000}{\text{normality}}$

Normality = $\frac{\text{given mass}}{\text{equivalent mass}}$

eq. mass = $\frac{\text{mol. mass}}{\text{valency}}.$

Note: Degree of dissociation (α) = $\frac{\Delta m - \text{given}}{\Delta m^{\circ} - \text{calculated}}$

$\Rightarrow$ Kohlraush's law

$$\Delta \bar{m} = v_+ \lambda_+^\infty + v_- \lambda_-^\infty$$

$\Rightarrow$ Faraday's law

$$\Rightarrow Q = I T \rightarrow \text{time (in secs)}$$

I Current

$$\Rightarrow m = Z Q \quad \text{Electrochemical Equivalent}$$

$$\Rightarrow w = Z I T / F \rightarrow 96500$$

$$\Rightarrow m = \frac{E I T}{n F}$$

⇒ Numericals Solutions

* Constants

i) $R = 8.314 \text{ J/mol/K}$

$F = 96500$

$T = 298 \text{ K}$

$$\text{Formulae: } = \left[E^{\circ} - \frac{2.303RT}{nF} \log_{10} \frac{[\text{pdt}]}{[\text{reactant}]} \right]$$

$$= \left[E^{\circ} - \frac{0.059}{n} \log_{10} \frac{[\text{pdt}]}{[\text{react}]} \right]$$

⇒ EMF Numericals

1) E° Cell from standard Electrode Potentials

Q.1 Calculate E° for the cell
~~Zn~~ $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$

$E^{\circ} (\text{Cu}^{2+}/\text{Cu}) = 0.34 \text{ V}$

$E^{\circ} (\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$

[Idea] Always the positive value or greater value is cathode

Thus,

$$E^{\circ} = E_c - E_a$$

$$E^{\circ} = 0.34 - (-0.76)$$

$$E^{\circ} = 1.10 \text{ V}$$

2.] Type 2 : Type 2 Nernst Equation.
under non-standard conditions

Question $\text{Zn} | \text{Zn}^{2+} (0.1\text{M}) || \text{Cu}^{2+} (1.0\text{M}) | \text{Cu}$

$$E^{\circ} = 1.10\text{V}$$

Calculate : EMF.

Answer: Write Equation.



$n=2$. (as both half reactions have 2).

$$a = \frac{pdt}{\text{reactant}}$$

$$Q = \frac{0.1}{1} = 0.1$$

$$\text{EMF} = 1.10 - \frac{0.0591}{2} \times \log(0.1)$$

$$[= 1.1296\text{V}.]$$

3] For the cell $\text{Ni} | \text{Ni}^{2+} (1.0\text{M}) || \text{Ag}^{+} (x\text{M}) | \text{Ag}$

$$E^{\circ}_{\text{cell}} = 1.05\text{V}$$

$$E^{\circ} (\text{Ag}^{+}/\text{Ag}) = 0.80\text{V}$$

$$E^{\circ} (\text{Ni}^{2+}/\text{Ni}) = -0.25\text{V}$$

$$\text{EMF} = 1.05\text{V}.$$

$n=2$.

$$Q = \frac{1}{[\text{Ag}^{+}]^2}$$

$$E_{\text{cell}} = E^{\circ} - \left(\frac{0.0591}{n} \right) \log \left(\frac{1}{[\text{Ag}^{+}]^2} \right)$$

$$1 = 1.09 - \left(\frac{0.0591}{2} \right) \log \left(\frac{1}{Ag^{+2}} \right)$$

$$0 = 1.092 \log [Ag^{+}]$$

$$= 10^{1-0.846}$$

$$= 0.143 M$$

4.

⇒ Type 4 : Concentration Cells

Question) Calculate E°_{cell} for the cell $Cu | Cu^{2+} (0.01M) || Cu^{2+} (1.0M) | Cu (s)$ at 298K.

Note: As this is a concentration cell $E^{\circ}_{cell} = 0$.
 (The electrode which is more dilute = 0.01M anode.
 concentrate = 1.0M cathode)

$$Q = \frac{0.01}{1} = 0.01$$

$$EMF = 0 - \left(\frac{0.0591}{2} \right) \log (0.01)$$

$$= - (0.02955) \times (-2)$$

$$= 0.0591 V$$

5) Type 5

⇒ Calculate the electrode potential of a Zn electrode dipped in 0.01 M $ZnSO_4$ soln. Given $E^\circ (Zn^{2+}/Zn) = -0.76V$.

$$n = 2$$

Idea

: For a single electrode the formula changes to reduction

$$EMF = E^\circ + \left(\frac{0.0591}{n} \right) \times \log(\text{ion conc.})$$

6) Type 6 :

⇒ The standard cell potential for a reaction is $E^\circ_{\text{cell}} = 0.295V$ with $n = 2$. Calculate equilibrium constant K_c at 298K.

Idea : At Equilibrium $E^\circ_{\text{cell}} = 0$ and $Q = K_c$.

$$0 = E^\circ - \left(\frac{0.0591}{n} \right) \log K_c.$$

$$\log K_c = \frac{2 \times 0.295}{0.0591}$$

$$\log K_c = 9.983$$

$$K_c = 9.62 \times 10^9$$

⇒ GIBBS FREE ENERGY NUMERICALS.

1) $\Delta G = \Delta H - T\Delta S$ (spontaneity and transition temperature)

Question) $\Delta H = -92.0 \text{ kJ/mol} = -92000 \text{ J/mol}$
 $\Delta S = -198.0 \text{ J/K mol}$
 $T = 298 \text{ K}$

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= -92000 - (298 \times (-198)) \\ &= -32996 \text{ J/mol} \\ &= \boxed{-3.2996 \text{ kJ/mol}}\end{aligned}$$

Note) To find the temp at which it will become non-spontaneous.

Idea: For that $\Delta G = 0$.

$$\begin{aligned}0 &= -92000 - T \times (-198) \\ \boxed{T &= 464.6 \text{ K}}\end{aligned}$$

2) Type 2: Calculate ΔG when K_c is given

$K_c = 10$
 $T = 298 \text{ K}$

$$\begin{aligned}\Delta G &= -2.303RT \log K_c \\ &= -2.303 \times 8.314 \times 298 \times \log(10) \\ &= -5705.8 \text{ J/mol} \\ &= \boxed{-5.705 \text{ kJ/mol}}\end{aligned}$$

3) ΔG and EMF (the bridge formula)

Question) $E^\circ = 1.10 \text{ V}$

$n = 2$

$\Delta G = ?$

$$\Delta G = -nFE^\circ$$

$$\Delta G = -2 \times 96500 \times 1.10$$

$$\Delta G = -212,300 \text{ J/mol}$$
$$= -212.3 \text{ kJ/mol}$$

4) Type 4 — ΔG (non-standard) using E_{cell}

$E^\circ = 1.1296 \text{ V}$

$n = 2$

$\Delta G = ?$

$$\Delta G = -nFE^\circ$$

$$\Delta G = -2 \times 96500 \times 1.1296 = -217900$$

$$\Delta G = -217.9 \text{ kJ/mol}$$